

Olist Business Intelligence Report DAX :

Measure 1:

Sales & Revenues:

- `Total Revenue = SUM(order_payments[payment_value])`
-
- `Total Orders = DISTINCTCOUNT(orders[order_id])`
-
- `Average Order Value = DIVIDE([Total Revenue], [Total Orders])`
-
- `Total Items Sold = COUNTROWS(order_items)`
-
- `Average Items per Order = DIVIDE([Total Items Sold], [Total Orders])`
-
- `Total Freight Cost = SUM(order_items[freight_value])`
-
- `Freight as % of Order Value = DIVIDE([Total Freight Cost], SUM(order_items[price]))`

Measure 2:

Time Intelligence

- `Revenue YTD = TOTALYTD([Total Revenue], 'Date'[Date])`
- `Revenue PY = CALCULATE([Total Revenue], SAMEPERIODLASTYEAR('Date'[Date]))`
- `Revenue YoY % = DIVIDE([Total Revenue] - [Revenue PY], [Revenue PY])`
- `Revenue MTD = TOTALMTD([Total Revenue], 'Date'[Date])`

- Revenue QTD = TOTALQTD([Total Revenue], 'Date'[Date])

```
-- Matched-cutoff YTD (important since 2018 data is partial)
Revenue YTD Matched =
CALCULATE(
    [Total Revenue],
    FILTER(
        ALL('Date'),
        'Date'[Year] = YEAR(MAX('Date'[Date])) &&
        'Date'[DayOfYear] <= DAY(MAX('Date'[Date]))
            + DAYS(0, MONTH(MAX('Date'[Date])) - 1, 0) -- use your
DayOfYear column instead if built
    )
)
```

- Rolling 3M Revenue =

```
CALCULATE([Total Revenue], DATESINPERIOD('Date'[Date],
LASTDATE('Date'[Date]), -3, MONTH))
```

Measure 3:

Delivery Performance:

- Avg Delivery Days =

```
AVERAGEX(
    orders,
    DATEDIFF(orders[order_purchase_timestamp],
orders[order_delivered_customer_date], DAY)
)
```

- Delivered Orders =

```
CALCULATE([Total Orders],
NOT(ISBLANK(orders[order_delivered_customer_date])))
```

On-Time Deliveries =

```
CALCULATE(
    [Total Orders],
    orders[order_delivered_customer_date] <=
orders[order_estimated_delivery_date]
)
```

- On-Time Delivery Rate = $\text{DIVIDE}([\text{On-Time Deliveries}], [\text{Delivered Orders}])$

- Avg Delay Days (Late Only) =
 $\text{CALCULATE}(\text{AVERAGEX}(\text{orders}, \text{DATEDIFF}(\text{orders}[\text{order_estimated_delivery_date}], \text{orders}[\text{order_delivered_customer_date}], \text{DAY})), \text{orders}[\text{order_delivered_customer_date}] > \text{orders}[\text{order_estimated_delivery_date}])$

Measure 4:

Customer Metrics:

- Total Unique Customers =
 $\text{DISTINCTCOUNT}(\text{customers}[\text{customer_unique_id}])$

Repeat Customers =

 $\text{CALCULATE}(\text{DISTINCTCOUNT}(\text{customers}[\text{customer_unique_id}]), \text{FILTER}(\text{VALUES}(\text{customers}[\text{customer_unique_id}]), \text{CALCULATE}(\text{DISTINCTCOUNT}(\text{orders}[\text{order_id}])) > 1))$

- Repeat Purchase Rate = $\text{DIVIDE}([\text{Repeat Customers}], [\text{Total Unique Customers}])$

- New Customers =
`CALCULATE(
 DISTINCTCOUNT(customers[customer_unique_id]),
 FILTER(
 VALUES(customers[customer_unique_id]),
 CALCULATE(MIN(orders[order_purchase_timestamp]))
 >= STARTOFMONTH(MAX('Date'[Date]))
)
)`

Measure 5 :
 Product / Category

- Revenue by Category =
`CALCULATE([Total Revenue], USERELATIONSHIP(order_items[product_id],
 products[product_id]))`
- Top Category Rank =
`RANKX(ALL(products[category_name_english]), [Total Revenue], , DESC)`
- Category YoY Growth % =
`VAR CurrYear = [Total Revenue]
 VAR PrevYear = CALCULATE([Total Revenue],
 SAMEPERIODLASTYEAR('Date'[Date]))
 RETURN DIVIDE(CurrYear - PrevYear, PrevYear)`

Measure 6 :
 Reviews / CX

- Average Review Score = `AVERAGE(order_reviews[review_score])`
- Review Count = `COUNTROWS(order_reviews)`
- Low Score Orders (1-2) =
`CALCULATE([Review Count], order_reviews[review_score] <= 2)`

- Low Score Rate = DIVIDE([Low Score Orders (1-2)], [Review Count])

-- Correlation-style measure: avg score for delayed vs on-time (build as two measures, compare in visual)

Avg Score - Delayed Orders =

```
CALCULATE(
    [Average Review Score],
    orders[order_delivered_customer_date] >
orders[order_estimated_delivery_date]
)
```

- Avg Score - OnTime Orders =

```
CALCULATE(
    [Average Review Score],
    orders[order_delivered_customer_date] <=
orders[order_estimated_delivery_date]
)
```

Measure 7:

Seller Performance:

- Revenue per Seller = [Total Revenue] -- sliced by seller_id in a table visual

- Seller Rank by Revenue =

```
RANKX(ALL(sellers[seller_id]), [Total Revenue], , DESC)
```

Seller On-Time Rate = [On-Time Delivery Rate] -- sliced by seller_id

Active Sellers = DISTINCTCOUNT(order_items[seller_id])

Measure 8 :

Cancellations / OPS

- Cancelled Orders =
CALCULATE([Total Orders], orders[order_status] = "canceled")
- Cancellation Rate = DIVIDE([Cancelled Orders], [Total Orders])
- Unavailable Orders =
CALCULATE([Total Orders], orders[order_status] = "unavailable")

**M - CODE TO CONVERT A DATE COL FROM 20210608 to Proper Date
format i.e. 08/06/2021**

```
Date.FromText(  
    Text.Range(Text.From([DueDateKey]), 0, 4) & "-" &  
    Text.Range(Text.From([DueDateKey]), 4, 2) & "-" &  
    Text.Range(Text.From([DueDateKey]), 6, 2)  
)
```

```
Date.FromText(  
    Text.Range(Text.From([ShipDateKey]), 0, 4) & "-" &  
    Text.Range(Text.From([ShipDateKey]), 4, 2) & "-" &  
    Text.Range(Text.From([ShipDateKey]), 6, 2)  
)
```

```
Date.FromText(  
    Text.Range(Text.From([ShipDateKey]), 0, 4) & "-" &  
    Text.Range(Text.From([ShipDateKey]), 4, 2) & "-" &  
    Text.Range(Text.From([ShipDateKey]), 6, 2)  
)
```

This DAX is used to build a time intelligence - DATE TABLE:

```
ADDCOLUMNS (
    CALENDAR (DATE (2017,1,1), DATE (2020,12,31)),
    "Year", YEAR([Date]),
    "Month Number", MONTH([Date]),
    "Day", DAY([Date]),
    "Month Name", FORMAT([Date], "MMM"),
    "Month Name Full", FORMAT([Date], "MMMM"),
    "Quarter", "Q" & FORMAT([Date], "Q"),
    "Year-Month", FORMAT([Date], "YYYY-MM"),
    "Year-Quarter", YEAR([Date]) & "-Q" & FORMAT([Date], "Q"),
    "Week Number", WEEKNUM([Date]),
    "Day of Week", FORMAT([Date], "ddd"),
    "Day of Week Full", FORMAT([Date], "dddd"),
    "Day Number (Mon=1)", WEEKDAY([Date], 2),
    "Is Weekend", WEEKDAY([Date], 2) > 5,
    "Day of Year", DATEDIFF (DATE (YEAR([Date]),1,1), [Date], DAY) + 1
)
```

```
"Year", YEAR([Date])           -- pulls just the year → 2017, 2018...
"Month Number", MONTH([Date])  -- pulls month as number → 1 to 12
"Day", DAY([Date])             -- pulls day of month → 1 to 31
```

Month Naming

```
Month Name = FORMAT([Date], "MMM")           --> "Jan", "Feb", "Mar"...
Month Name Full = FORMAT([Date], "MMMM")      --> "January", "February"...

Day Name = FORMAT([Date], "ddd")              --> "Mon", "Tue", "Wed"...
Day Name Full = FORMAT([Date], "dddd")        --> "Monday", "Tuesday"...
```